

EcoRAG

Experiment C

Evidence Sufficiency & Citation Correctness Evaluation

Generated: October 04, 2026

Dataset: 27 Environmental & Sustainability Questions

Model: Qwen2.5-3B-Instruct (CPU, bf16)

Embeddings: BGE bge-small-en-v1.5 (384 dim)

Review: Single-reviewer manual annotation

Executive Summary

Key Finding: EcoRAG demonstrates consistent answer generation with strong evidence utilization (91% coverage of available sources) but significant weaknesses in calibration (70% accuracy) and citation correctness (24% of claims incorrectly attributed). The system reliably answers when evidence is sufficient (100% accuracy, 14/14 questions) but fails to abstain when evidence is insufficient (44% abstention recall, 4/9 questions).

Metric	Result
Total Questions	27 (frozen)
Questions with Sufficient Evidence	14/27 (52%)
Overall Calibration Accuracy	19/27 (70%)
Answer Content Fully Correct	18/27 (67%)
Citations Incorrect (Manual)	20/85 (24%)
Evidence Coverage in Top-5	32/35 (91%)
Reviewers	1 (single-reviewer)

1. Overview & Methodology

Objective:

Measure EcoRAG's ability to retrieve evidence, generate grounded answers, and provide correct citations.

Question Set (n=27):

Type	Category	Count	Purpose
a	Single-chunk questions	6	Highest retrievability
b	Multi-chunk questions	5	Requires synthesis
c	Partially answerable	5	Incomplete evidence
d	Unanswerable	5	No relevant evidence
e	Dependent-source	6	Cross-document dependency

2. Evidence Sufficiency Metrics (S1–S6)

S1: Retrieval Sufficiency

Sufficiency Level	Count	Percentage
Sufficient	14	52%
Partial	4	15%
Insufficient	9	33%
Unit Retrieval Recall	35/45	78%

S2: Answer-Unit Coverage

Metric	Pooled	Per-Question Avg
S2-gen (in top-5)	32/35 = 91%	0.89
S2-e2e (all required)	35/55 = 64%	0.62

S3: Conclusion Justification

Metric	Distribution	Mean Score
S3a (cited passages)	10 just / 7 part / 5 not (n=22)	0.61
S3b (full top-5)	14 just / 3 part / 5 not (n=22)	0.70

S4: Calibrated Behaviour

Condition	Count	Accuracy
Sufficient → FULL	14/14	100%
Partial → PARTIAL+GAP	1/4	25%
Insufficient → ABSTAIN	4/9	44%
Overall Calibration	19/27	70%

3. Citation Correctness

Verifier Results:

Metric	Count	Rate
Claims with citations	81/82	99%
Misattributed	10/81	12%
Potentially supported	68/81	84%

Manual Review Results:

Status	Count	Rate
Supported	65/85	76%
Misattributed	11/85	13%
Unsupported	9/85	11%
Incorrect Total	20/85	24%

4. Failure Taxonomy

Failure Category	Units	Percentage
Retrieval Failure	10	18%
Calibration/Sufficiency	4	7%
Generation Failure	3	5%
Citation Failure	5	9%
Correct	33	60%

5. Results by Evidence Condition

Sufficient Evidence (14 questions):

Metric	Value
Calibration (FULL)	14/14 = 100%
Content fully correct	13/14 = 93%
All citations correct	7/14 = 50%

Insufficient Evidence (9 questions):

Metric	Value
Abstention recall	4/9 = 44%
Over-answered	5/9 = 56%
Content fully correct	4/9 = 44%

6. Results by Question Type

Type	Calib. Acc.	Content OK	Cit. Correct	Finding
a (single-chunk)	83%	83%	83%	Excellent
b (multi-chunk)	80%	60%	20%	Citations weak
c (partial)	20%	20%	20%	Hardest
d (unanswerable)	80%	80%	80%	Good abstention
e (dependent)	83%	83%	17%	Citations broken

7. What This Experiment Tells Us

- 1. Strong evidence utilization.** When required units are in top-5, EcoRAG answers correctly 100% of the time. It covers 91% of available sources and justifies 70% of claims.
- 2. Calibration failure.** When evidence is insufficient, the model should abstain. It does so only 44% of the time, answering anyway by substituting related entities or generalizing beyond scope.
- 3. Citation errors.** 24% of cited claims incorrectly point to sources. This is especially severe in dependent-source questions (78% incorrect).
- 4. Retrieval is secondary.** 18% of required units are not retrieved. The bigger problem is over-answering when evidence is absent.
- 5. Single reviewer, frozen rubric.** 27 questions with 3 post-hoc corrections found. Results are lower bound on agreement.

Appendix: Source Documents

9 PDFs, 1,242 chunks:

Document	Chunks	Topic
GMSR 2025	~500	Global Methane & Sustainability
Mapping Green Jobs	~200	Energy Job Mapping
Modernising Grids	~180	Grid Infrastructure
Managing Seasonal Electricity	~130	Demand/Supply
Critical Minerals	~80	Norway Minerals
Other	~152	Finance, standards, policy

Report generated: 2026-10-04 13:52 UTC
Design: Frozen before run; single reviewer; manual review post-hoc.